

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Computer Science and Engineering
B.Tech II Semester-III
CS231 – Object Oriented Programming Practical Assignment
Lab Assignment – 5

1. Write a C++ program : Digital Wallet Transfer

Create a class *DigitalWallet* with:

- *walletID* (int), *ownerName* (string), and *balance* (double).

Requirements:

- Use **default and parameterized constructors**.
- Write a **friend function** *sendMoney(DigitalWallet &sender, DigitalWallet &receiver, double amount)* that transfers money from one wallet to another and returns the updated sender wallet object.
- Use a **const member function** *showBalance()* to display wallet details.
- Ensure **balance cannot go negative** (print an error if funds are insufficient).

Sample Input :

Wallet1: 201, Alex, 1200

Wallet2: 202, Sam, 800

Transfer 500 from Alex to Sam

Sample Output :

After Transfer:

Alex: Balance = 700

Sam: Balance = 1300

2. Write a C++ program to Video Duration Merger

Create a class *VideoDuration* to represent the duration of a video in hours, minutes, and seconds.

Requirements:

- Provide **default and parameterized constructors**.
- Write a **friend function** *mergeDuration(VideoDuration v1, VideoDuration v2)* that combines the lengths of two videos and returns a new *VideoDuration* object.
- Handle proper carry-over (60 seconds = 1 minute, 60 minutes = 1 hour).
- Use a **const member function** *display()* to show the result in *hh:mm:ss* format.

Sample Input :

Video 1: 1:45:20

Video 2: 0:35:50

Sample Output :

Total Duration: 2:21:10

3. Write a program using function overloading to perform some arithmetic operations on different types of arguments.
4. Write a program to compute the area of square, rectangle and circle by overloading the *area()* function.
5. Write a function *power()* to raise a number *m* to a power *n*. The function takes a double value for *m* and integer value for *n*, and returns the result correctly. Use a default value of 2 for *n* to make the function to calculate square when this argument is omitted. Write a main function that gets the values of *m* and *n* from the user to test the function. Use the concept of function overloading.
6. Write a C++ program to create a class named *Box* that has:
- One private data member: *length* (integer).
 - A constructor to initialize the length.
- (a) Declare a friend function named *displayLength(Box)* that:
- Can access the private member *length* of the *Box* class.
 - Displays the length of the box.
- (b) Define an inline function named *square(int)* that:
- Returns the square of the number passed to it.
 - Demonstrate its use in the *main()* function.
7. Write a C++ program to create two classes, *ClassA* and *ClassB*, each having a private integer member. Write a friend function *add()* that:
- Takes objects of both classes as arguments.
 - Accesses their private data.
 - Returns the sum of the two numbers.
8. Write a C++ program to swap the values of two numbers by using

- Call by value
- Call by address
- Call by reference variable

9. Write a C++ program to overload a function fun() as follows:

- Case 1: void fun(const int i)
void fun(int i)
- Case 2: void fun(char *a)
void fun(const char *a)

10. Write a C++ program to find average of two numbers and consider following cases:

- Case 1: Function with no argument and no return value- void avg()
- Case 2: Function with no argument but with return value- float avg()
- Case 3: Function with argument and but no return value- void avg(float, float)
- Case 4: Function with argument and return value- float avg(float, float)